

ME381 ROBOTICS

Experiment No.: 4

Experiment Title: Actuators: DC motor with encoder

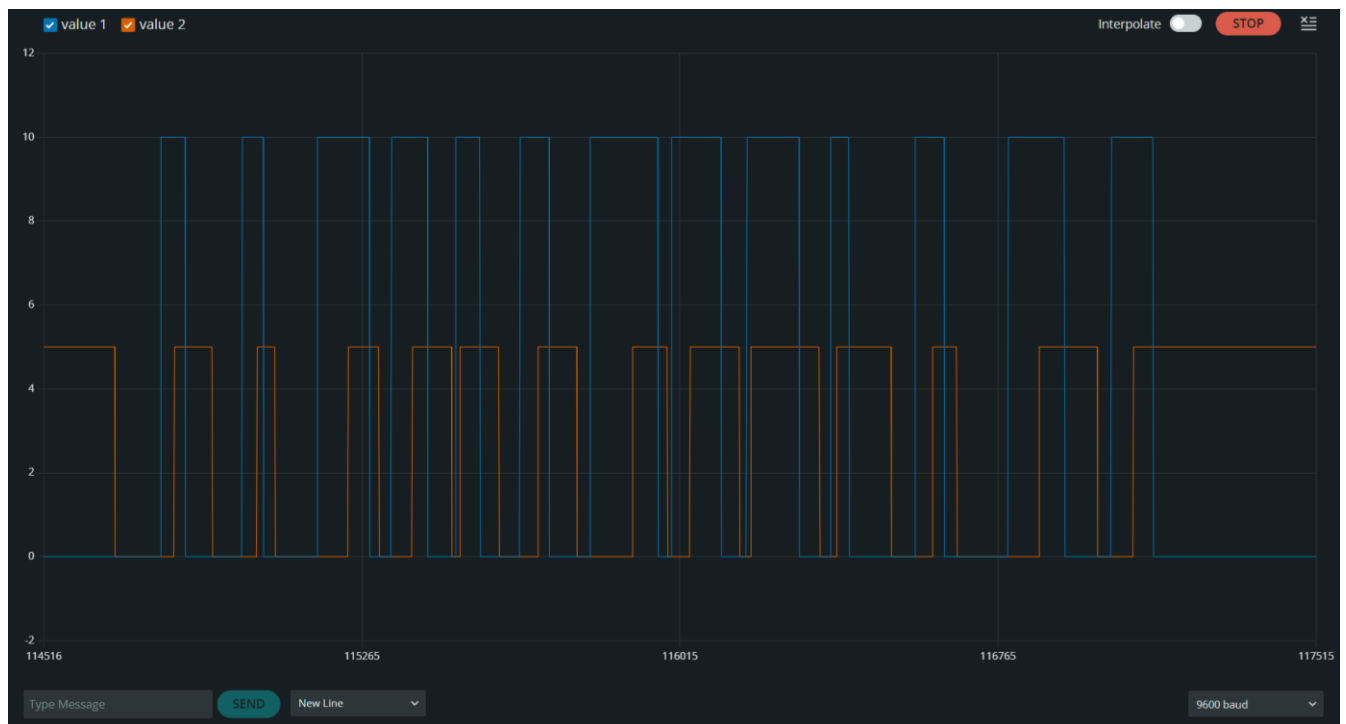
Group No.: B6

Name: Pritam Priyadarshi

Roll no.: 230793

Results for the group task:

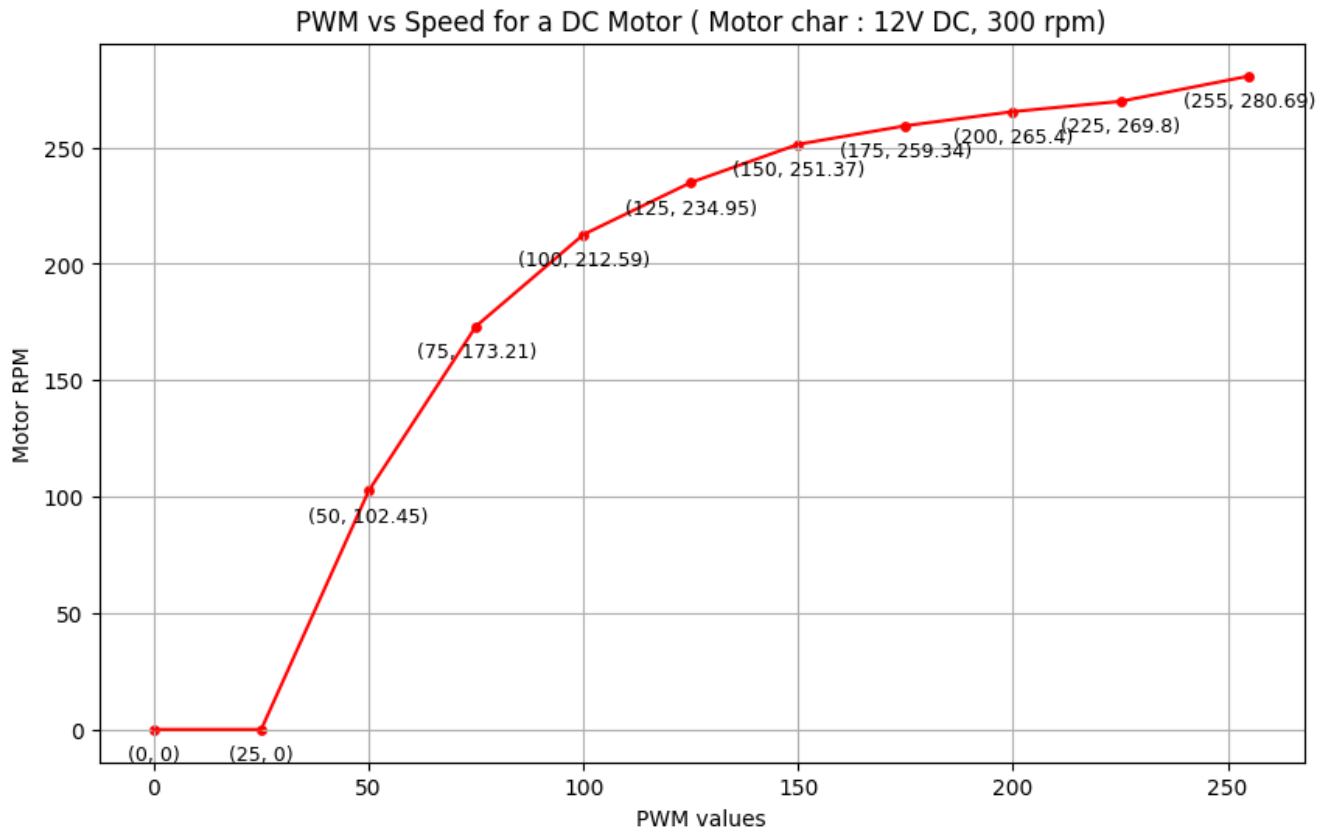
1.



- Number of pulses of A and B per rotation is 13 and 12 respectively.
- When the encoder is seen from the back and rotated CW, red leads blue.

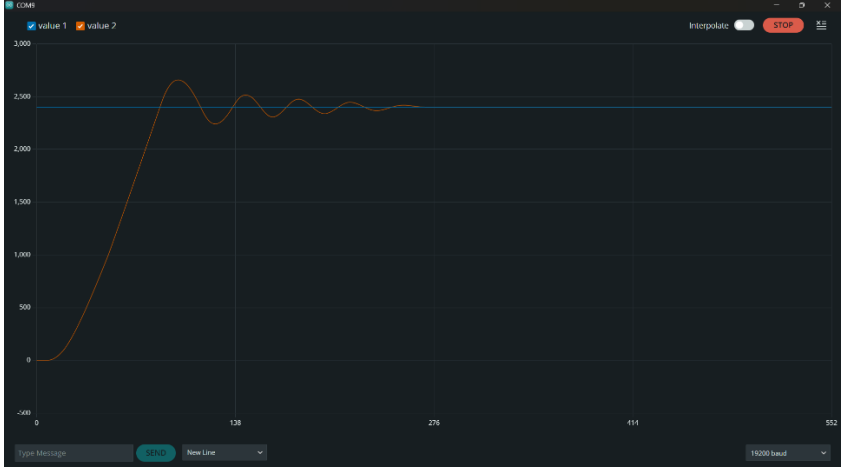
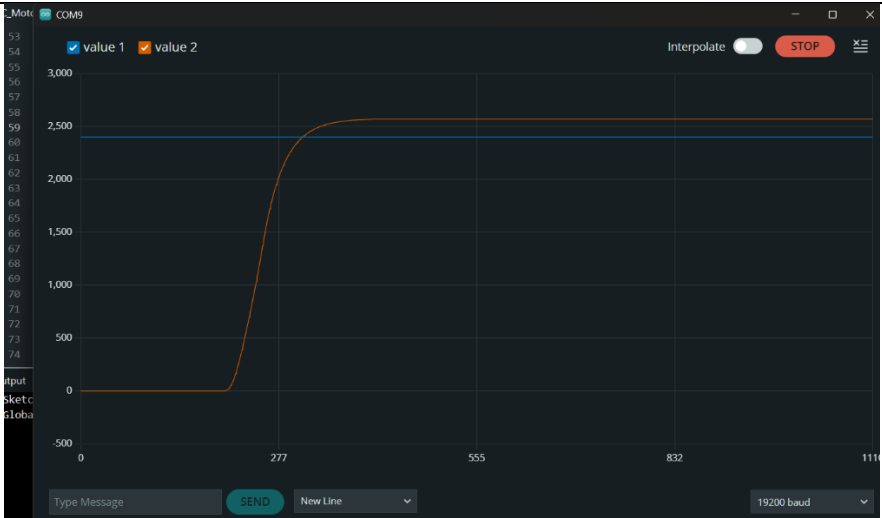
ME381 ROBOTICS

2.

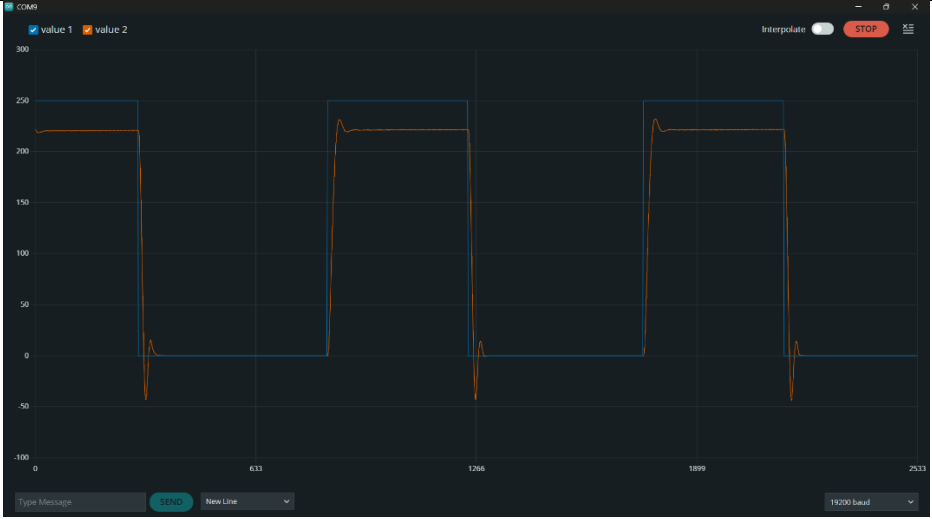
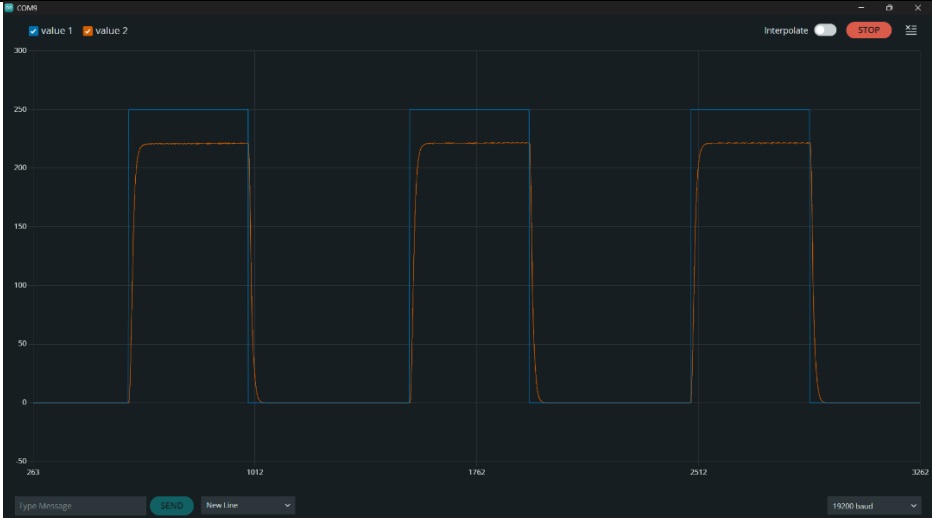
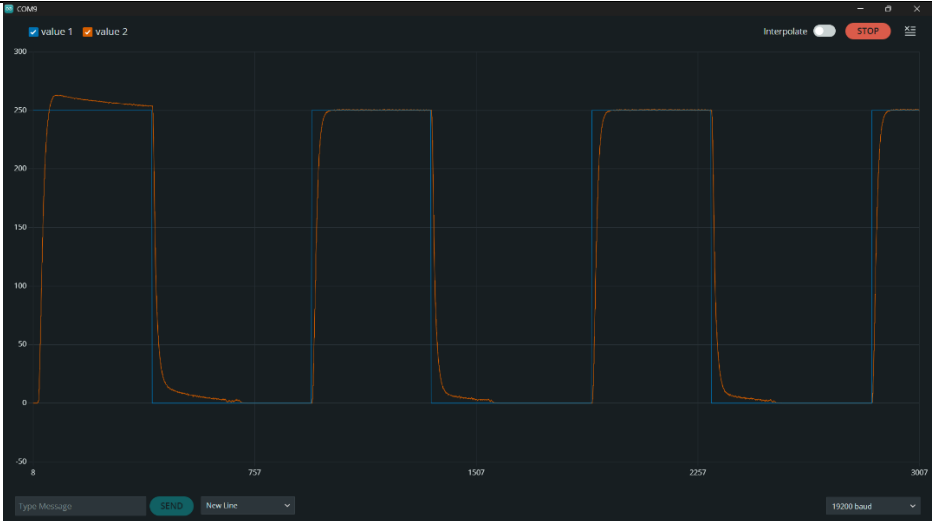


ME381 ROBOTICS

3.

Target	P	I	D	Image of the response curve in serial plotter
Position Control	3	0	0	
	3	0	0.5	
	3	0.05	0.5	

ME381 ROBOTICS

Target	P	I	D	Image of the response curve in serial plotter
Velocity control	4	0	0	
	4	0	0.2	
	4	5	0.2	

ME381 ROBOTICS

Feedback on this experiment

The experiment gave a good understanding of actuator behavior with encoders. Observations on pulse counts and PID tuning enhanced clarity of control response. Also it successfully demonstrated encoder functionality and PID response in both position and velocity control.